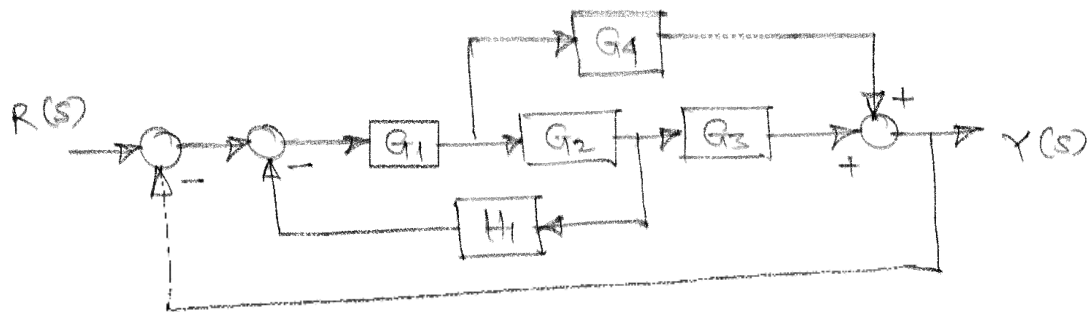
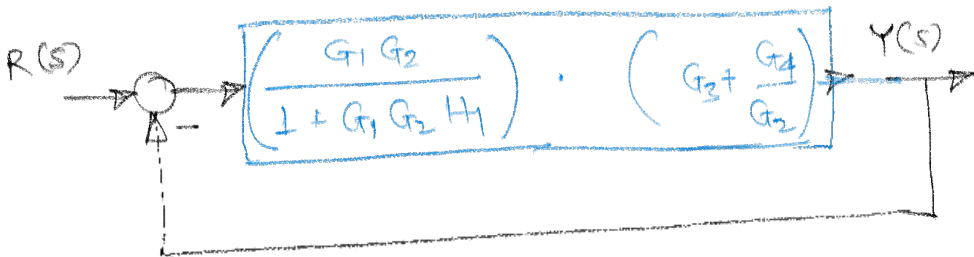


Ex.

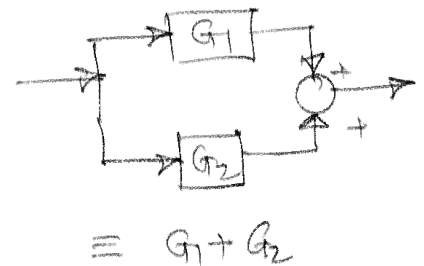
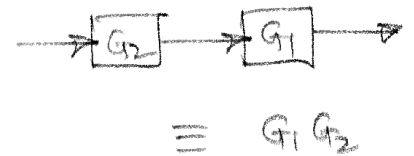
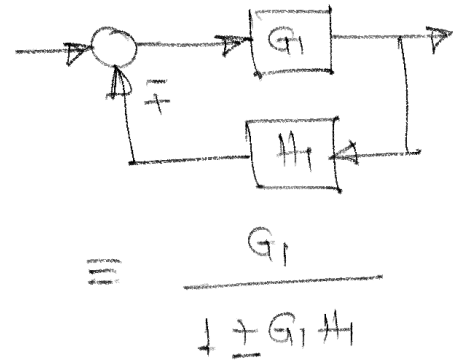


Find $\frac{Y(s)}{R(s)}$.



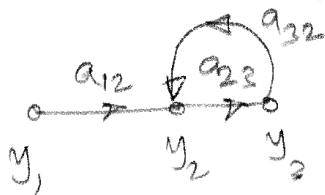
$$\frac{Y(s)}{R(s)} = \frac{\left(\frac{G_1 G_2}{1 + G_1 G_2 H_1} \right) \cdot \left(G_3 + \frac{G_4}{G_2} \right)}{1 + \frac{G_1 G_2}{1 + G_1 G_2 H_1} \cdot \left(G_3 + \frac{G_4}{G_2} \right)}$$

$$= \frac{G_1 G_2 G_3 + G_1 G_4}{1 + G_1 G_2 H_1 + G_1 G_2 G_3 + G_1 G_4}$$



Signal flow graph (SFG)

Introduced by S-J. Mason

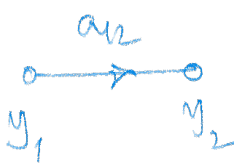


$$y_2 = a_{12} y_1 + a_{32} y_3, \quad a_{ij} \in \mathbb{R}.$$

$$y_3 = a_{23} y_2$$

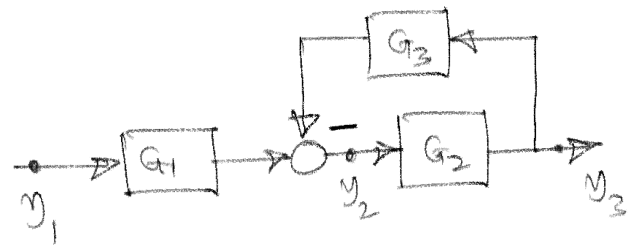
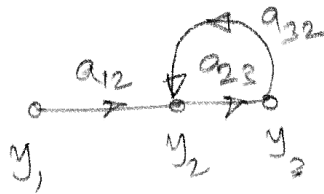
↑ gains.

y_1 : input node, y_3 = output node.



$$y_2 = a_{12} y_1$$

$$\boxed{y_1 = a_{12}^{-1} y_2} \leftarrow \text{not allowed.}$$

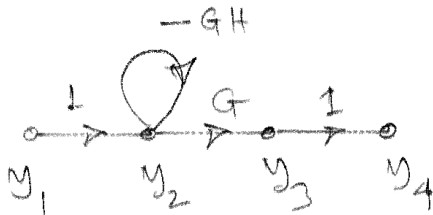


$$a_{32} \equiv -G_3$$

$$a_{23} \equiv G_2$$

$$a_{12} \equiv G_1$$

Ex.

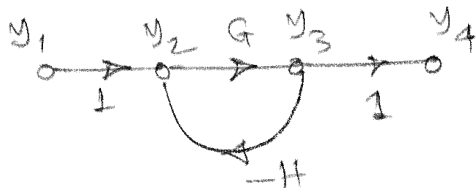


$$y_2 = y_1 - GH y_2 = y_1 - H y_3$$

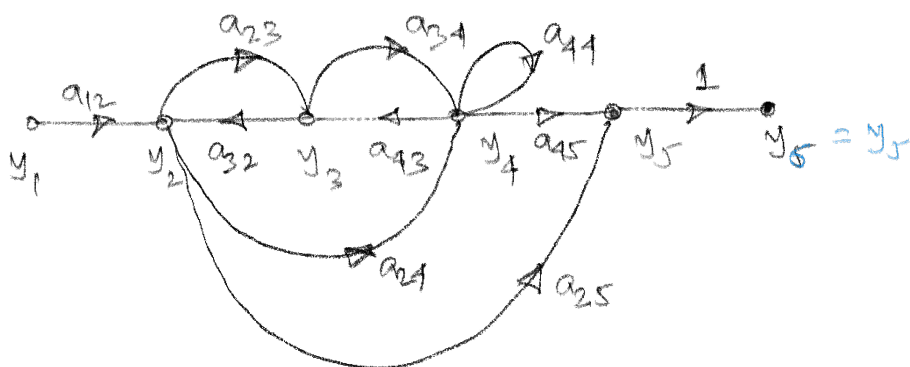
$$y_2 = \frac{1}{1+GH} y_1$$

$$y_3 = G y_2 = \frac{G}{1+GH} y_1$$

|||



Ex.

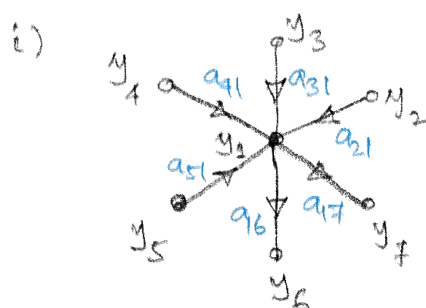


- **Input node**: it has only outgoing branches
- **Output node**: " " incoming "
- **Path**: is any collection of a continuous succession of branches traversed in the same dir?
- **Forward path**: a path that starts at an input node and ends at an output node, and along

which no node is traversed more than once.

- **Loop**: a path that originates and terminates on the same node and along which no other node is encountered more than once
- **Non-touching loops**: two (or more) loops are non-touching if they do not share a common node.
- **Path gain**: the product of the branch gains encountered in traversing a path
e.g. Path: $y_1 - y_2 - y_3 - y_4 - y_5$; path gain: $a_{12}a_{23}a_{34}a_{45}$.

* SFG algebra:



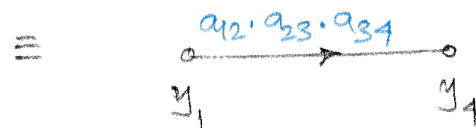
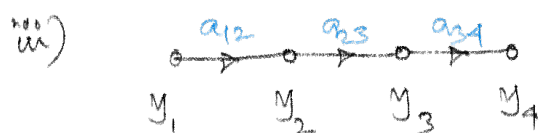
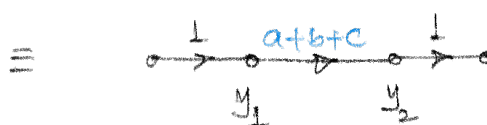
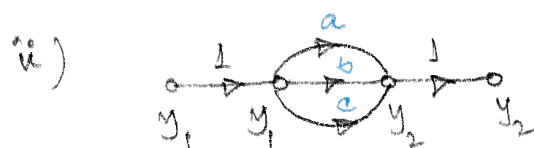
$$y_1 = a_{21}y_2 + a_{31}y_3 + a_{41}y_4 + a_{51}y_5$$

$$y_6 = a_{16}y_1$$

$$y_7 = a_{17}y_1$$

Input nodes: y_2, y_3, y_4, y_5 ,

Output nodes: y_1, y_6, y_7 .



* Mason's gain formula:

Overall gain from the input node y_{in} to the output node y_{out} can be written as

$$M = \frac{y_{out}}{y_{in}} = \sum_{k=1}^N \frac{M_k \Delta_k}{\Delta}$$

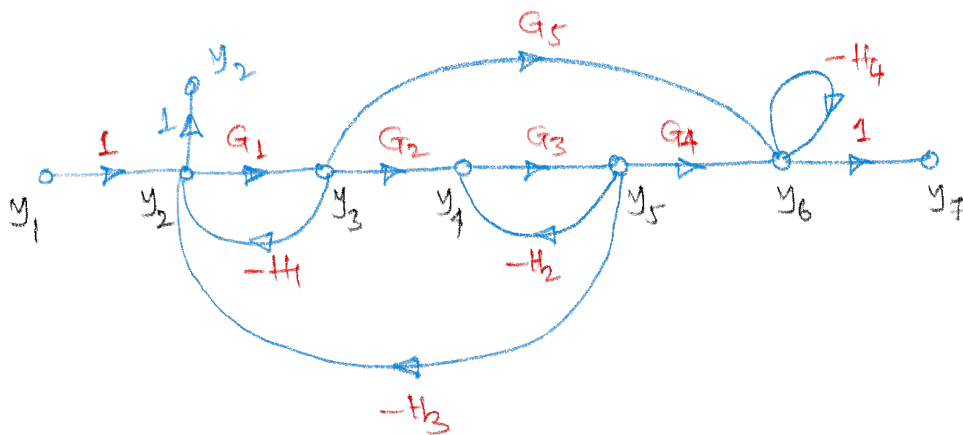
where, N is the total number of forward paths between y_{in} and y_{out} ,

M_k is the gain of the k^{th} forward path

$$\Delta := 1 - \left(\begin{aligned} &\text{sum of the gains of all individual loops} \\ &+ \left(\text{sum of the products of gains of all possible} \right. \\ &\quad \left. \text{combinations of two non-touching loops} \right) \\ &- \left(\text{sum of the products of gains of all possible} \right. \\ &\quad \left. \text{combinations of three non-touching loops} \right) \\ &+ \dots \end{aligned} \right)$$

Δ_k is the Δ for that part of the graph that is non-touching with the k^{th} forward path.

Ex.



Find $\frac{y_7}{y_1}$.

Find $\frac{y_2}{y_1}$.

[Exercise]

forward paths: i) $y_1 - y_2 - y_3 - y_4 - y_5 - y_6 - y_7$

ii) $y_1 - y_2 - y_3 - y_6 - y_7$

forward path gains: i) $\underbrace{G_1 G_2 G_3 G_4}_{M_1}$, ii) $\underbrace{G_1 G_5}_{M_2}$

$$\Delta = 1 - \left(\begin{aligned} &-G_1 H_1 - G_3 H_2 - H_4 - G_1 G_2 G_3 H_3 \\ &+ (G_1 H_1 G_3 H_2 + G_1 H_1 H_4 + G_3 H_2 H_4 + G_1 G_2 G_3 H_3 H_4) \\ &- (-G_1 H_1 G_3 H_2 H_4) \end{aligned} \right)$$

$$\Delta_1 = 1$$

$$\Delta_2 = 1 - (-G_3 H_2)$$

$$\frac{y_7}{y_1} = \frac{G_1 G_2 G_3 G_4 + G_1 G_5 (1 + G_3 H_2)}{1 - (-G_1 H_1 - G_3 H_2 - H_4 - G_1 G_2 G_3 H_2) + (G_1 H_1 G_3 H_2 + G_1 H_1 H_4 + G_3 H_2 H_4 + G_1 G_2 G_3 H_2 H_4) - (-G_1 H_1 G_3 H_2 H_4)}$$

$$= \frac{G_1 G_2 G_3 G_4 + G_1 G_5 (1 + G_3 H_2)}{1 + G_1 H_1 + G_3 H_2 + G_1 G_2 G_3 H_2 + H_4 + G_1 H_1 G_3 H_2 + G_3 H_2 H_4 + G_1 H_1 H_4 + G_1 G_2 G_3 H_2 H_4 + G_1 H_1 G_3 H_2 H_4}$$